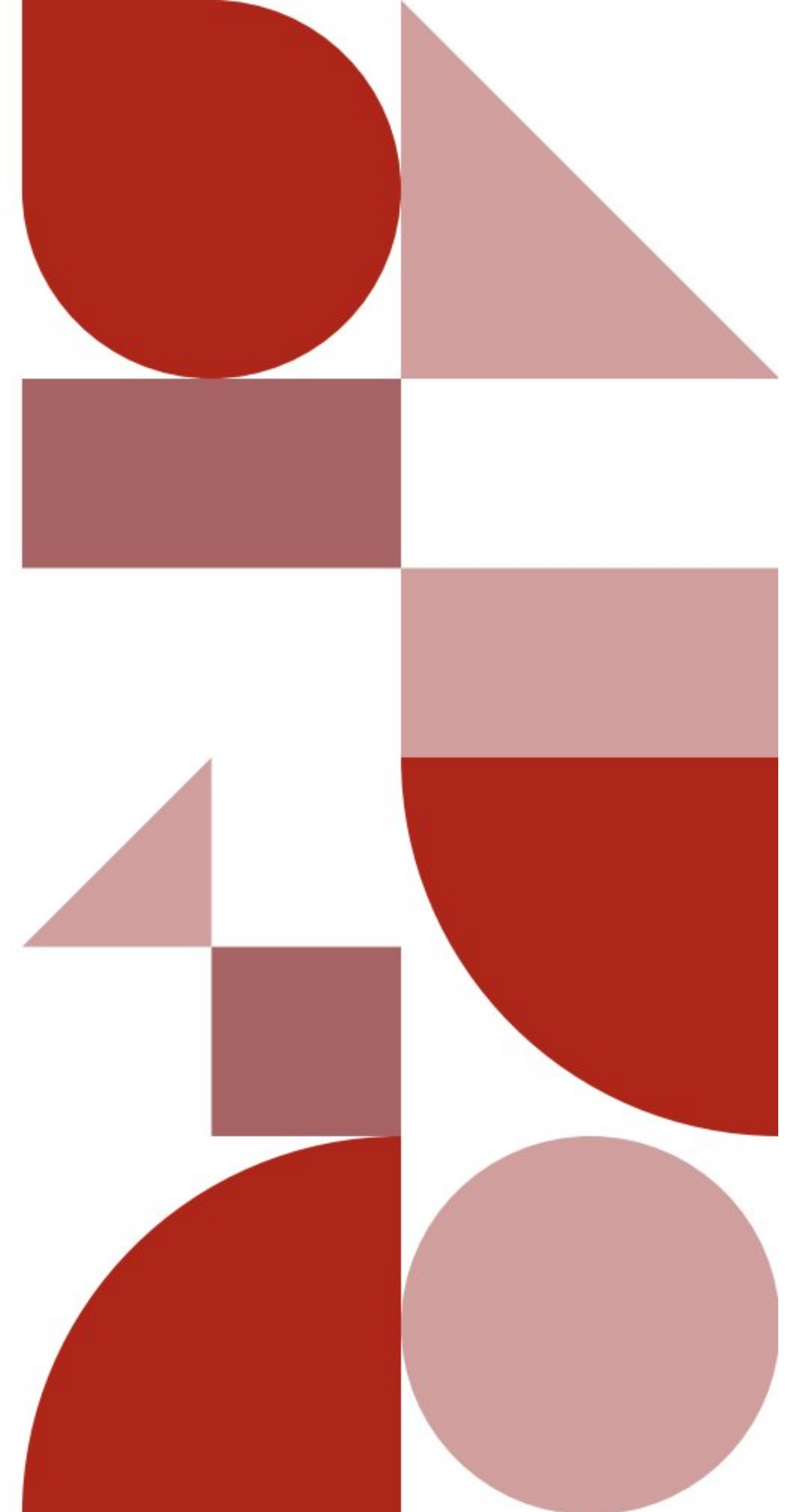




WEEK 8

CS50 INTRODUCTION INTO COMPUTER SCIENCE



JAVASCRIPT

Objects in JavaScript

JAVASCRIPT

1- Object Declaration & Properties

```
JS script.js > ...
1  let person = {
2      name: "Alice",
3      age: 25,
4      greet: function() {
5          console.log("Hello, my name is " + this.name);
6      }
7  };
8
9  // Accessing properties
10 console.log(person.name);
11 console.log(person["age"]);
12
13 // Calling a method
14 person.greet();
```

2- Modifying Properties

```
17 let book = {
18     title: "JavaScript Basics",
19     author: "John Doe"
20 };
21
22 // Add a new property
23 book.year = 2024;
24
25 // Update an existing property
26 book.title = "Advanced JavaScript";
27
28 // Delete a property
29 delete book.author;
30
31 console.log(book);
32 // Output: { title: 'Advanced JavaScript', year: 2024 }
```


JAVASCRIPT

3- Function inside Object

```
36 let calculator = {
37     add: function(a, b) {
38         return a + b;
39     },
40     subtract: function(a, b) {
41         return a - b;
42     }
43 };
44
45 console.log(calculator.add(5, 3)); // Output: 8
46 console.log(calculator.subtract(5, 3)); // Output: 2
47
```

4- Object inside Object

```
50 let company = {
51     name: "TechCorp",
52     employees: [
53         { name: "Alice", role: "Developer" },
54         { name: "Bob", role: "Designer" }
55     ]
56 };
57
58 console.log(company.employees[0].name); // Output: Alice
59
```

JAVASCRIPT

Arrays & Strings

JAVASCRIPT

2- Array Methods

1- Array Declaration

```
JS script.js > ...
1  array
2
3  let fruits = ["apple", "banana", "cherry"];
4  console.log(fruits[0]); // Output: apple
5  console.log(fruits[2]); // Output: cherry
6
```

```
7  array built-in method
8
9  fruits.push("orange"); // Add to the end
10 console.log(fruits); // ["apple", "banana", "cherry", "orange"]
11
12 fruits.pop(); // Remove from the end
13 console.log(fruits); // ["apple", "banana", "cherry"]
14
15 fruits.unshift("grape"); // Add to the beginning
16 console.log(fruits); // ["grape", "apple", "banana", "cherry"]
17
18 fruits.shift(); // Remove from the beginning
19 console.log(fruits); // ["apple", "banana", "cherry"]
20
21
22 ✓ fruits.forEach((fruit) => {
23   |   console.log(fruit);
24   | }); // Output: apple, banana, cherry
25
26
27 fruits.sort();
28 console.log(fruits); // Alphabetically sorted
29
30 fruits.reverse();
31 console.log(fruits); // Reversed order
32
33 console.log(fruits.indexOf("banana")); //index of elements Output: 1
34 fruits.splice(1, 1, "blueberry"); // Replace "banana" with "blueberry"
35 console.log(fruits); // ["apple", "blueberry", "cherry"]
36
```


JAVASCRIPT

3- String Declaration

```
38  string
39
40  let singleQuote = 'Hello';
41  let doubleQuote = "World";
42  let templateLiteral = singleQuote + " " + doubleQuote;
43  console.log(templateLiteral); // Output: Hello World
44
45
46  let greeting = "Hello";
47  console.log(greeting[1]); // Output: H
48
```

4- String Methods

```
49  string built-in method
50
51  console.log(greeting.charAt(4)); // Output: o
52
53  console.log(greeting.length); // Output: 5
54
55  console.log(greeting.toUpperCase()); // Output: HELLO
56
57  console.log(greeting.toLowerCase()); // Output: hello
58
59  let text = "I love JavaScript";
60  console.log(text.replace("JavaScript", "coding")); // Output: I love coding
61
62  let sentence = "JavaScript is fun!";
63  let words = sentence.split(" ");
64  console.log(words); // Output: ["JavaScript", "is", "fun!"]
65
66  let text = "JavaScript is fun";
67  console.log(text.endsWith("is", 14)); // Output: true (considers only the first 13 characters)
```

JAVASCRIPT

The DOM (Document Object Model)

JAVASCRIPT

1- QuerySelector

```
<> index.html > html > head
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  |   <title>querySelector Example</title>
5  </head>
6  <body>
7  |   <h1 id="title">Welcome to JavaScript!</h1>
8  |   <p class="info">This is the first paragraph.</p>
9  |   <p class="info">This is the second paragraph.</p>
10 |
11 |   <script src="script.js"></script>
12 </body>
13 </html>
14
```

```
1
2  let title = document.querySelector("#title"); // Select element by ID
3  let firstParagraph = document.querySelector(".info"); // Select first element with class "info"
4
5  console.log(title.textContent); // Output: Welcome to JavaScript!
6  console.log(firstParagraph.textContent); // Output: This is the first paragraph.
```


JAVASCRIPT

2- QuerySelectorAll

```
17 <!DOCTYPE html>
18 <html lang="en">
19 <head>
20   <title>querySelectorAll Example</title>
21 </head>
22 <body>
23   <h1>Article</h1>
24   <p class="info">First paragraph.</p>
25   <p class="info">Second paragraph.</p>
26   <p class="info">Third paragraph.</p>
27
28   <script src="script.js"></script>
29 </body>
30 </html>
31
```

OUTPUT IN CONSOLE :

Paragraph 1 : First paragraph

Paragraph 2 : Second paragraph

Paragraph 3 : Third paragraph

```
11 let paragraphs = document.querySelectorAll(".info"); // Select all elements with class "info"
12
13 // Iterate through the NodeList
14 paragraphs.forEach((para, index) => {
15   console.log("Paragraph " + (index + 1) + ": " + para.textContent);
16 });
```


JAVASCRIPT

3- innerHTML

```
34 <!DOCTYPE html>
35 <html lang="en">
36 <head>
37 |   <title>innerHTML Example</title>
38 </head>
39 <body>
40 |   <div id="container">Original Content</div>
41 |   <button onclick="changeContent()">Change Content</button>
42 |
43 |   <script src="script.js"></script>
44 </body>
45 </html>
46
```

OUTPUT IN HTML :
(Before Clicking button)

Original Content

OUTPUT IN HTML :
(After Clicking button)

New Content !

```
19
20 |   function changeContent() {
21 |       let container = document.querySelector("#container");
22 |       container.innerHTML = "<strong>New Content!</strong>";
23 |   }
```